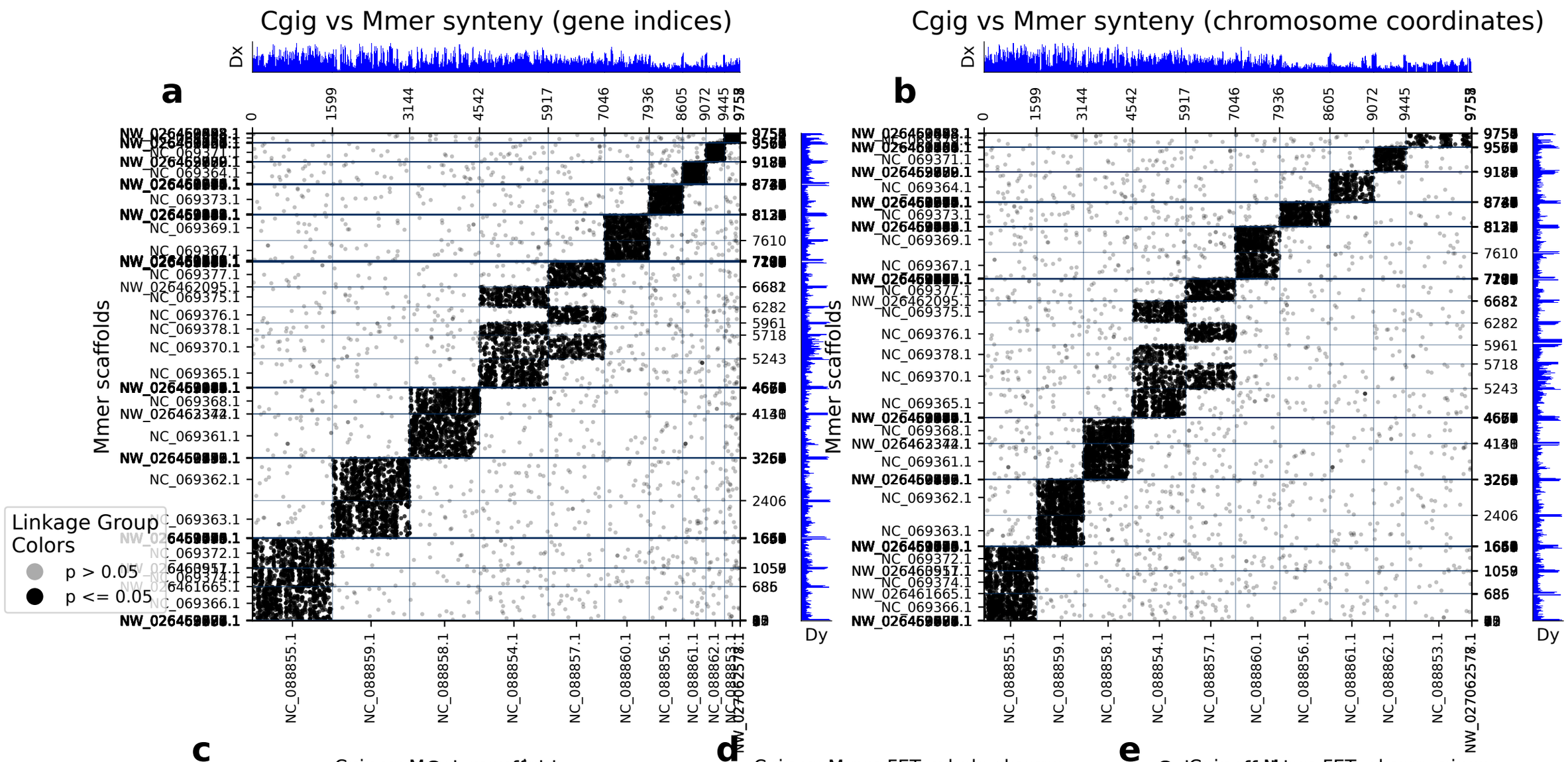
diamond: 9759 orthologs on 12 Cgig scaffolds and 182 Mmer scaffolds.



c

Cgig vs Mmer scaffolds

d

Cgig vs Mmer FET, whole chroms.

	Cgig scaf	Mmer scaf	p (FET)	Count
0	NC_088858.1	NC_069361.1	0.00e+00	750
1	NC_088859.1	NC_069362.1	0.00e+00	731
2	NC_088859.1	NC_069363.1	0.00e+00	639
3	NC_088855.1	NC_069366.1	0.00e+00	571
4	NC_088855.1	NC_069372.1	1.68e-310	494
5	NC_088856.1	NC_069373.1	0.00e+00	488
6	NC_088854.1	NC_069365.1	1.03e-310	462
7	NC_088858.1	NC_069368.1	0.00e+00	450
8	NC_088857.1	NC_069377.1	0.00e+00	425
9	NC_088860.1	NC_069369.1	0.00e+00	418
10	NC_088854.1	NC_069375.1	4.24e-221	331
11	NC_088861.1	NC_069364.1	0.00e+00	329
12	NC_088860.1	NC_069367.1	1.44e-266	316
13	NC_088855.1	NC_069374.1	2.50e-168	298
14	NC_088862.1	NC_069371.1	0.00e+00	283
15	NC_088857.1	NC_069376.1	9.39e-193	262
16	NC_088857.1	NC_069370.1	1.26e-93	232
17	NC_088854.1	NC_069378.1	2.81e-102	182
18	NC_088854.1	NC_069370.1	5.23e-31	170
19	NC_088853.1	NC_069379.1	9.94e-156	127

e

Cgig vs Mmer FET, chrom. pieces

	Cgig scaf	Mmer scaf	p (FET)	Count
0	NC_088858.1 : 0-end	NC_069361.1 : 0-end	0.00e+00	750
1	NC_088859.1 : 0-end	NC_069362.1 : 0-end	0.00e+00	731
2	NC_088859.1 : 0-end	NC_069363.1 : 0-end	0.00e+00	639
3	NC_088855.1 : 0-end	NC_069366.1 : 0-end	0.00e+00	571
4	NC_088855.1 : 0-end	NC_069372.1 : 0-end	1.68e-310	494
5	NC_088856.1 : 0-end	NC_069373.1 : 0-end	0.00e+00	488
6	NC_088854.1 : 0-end	NC_069365.1 : 0-end	1.03e-310	462
7	NC_088858.1 : 0-end	NC_069368.1 : 0-end	0.00e+00	450
8	NC_088857.1 : 0-end	NC_069377.1 : 0-end	0.00e+00	425
9	NC_088860.1 : 0-end	NC_069369.1 : 0-end	0.00e+00	418
10	NC_088854.1 : 0-end	NC_069375.1 : 0-end	4.24e-221	331
11	NC_088861.1 : 0-end	NC_069364.1 : 0-end	0.00e+00	329
12	NC_088860.1 : 0-end	NC_069367.1 : 0-end	1.44e-266	316
13	NC_088855.1 : 0-end	NC_069374.1 : 0-end	2.50e-168	298
14	NC_088862.1 : 0-end	NC_069371.1 : 0-end	0.00e+00	283
15	NC_088857.1 : 0-end	NC_069376.1 : 0-end	9.39e-193	262
16	NC_088857.1 : 0-end	NC_069370.1 : 0-end	1.26e-93	232
17	NC_088854.1 : 0-end	NC_069378.1 : 0-end	2.81e-102	182
18	NC_088854.1 : 0-end	NC_069370.1 : 0-end	5.23e-31	170
19	NC_088853.1 : 0-end	NC_069379.1 : 0-end	9.94e-156	127

(a) depicts the chromosomes/scaffolds of Cgig (x-axis) plotted against the chromosomes/scaffolds of Mmer (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 9759 orthologs found between 12 Cgig scaffolds and 182 Mmer scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chormosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.